MIL-A-3562B
2 OCTOBER 1959
SUPERSEDING
MIL-S-003562A(CmlC)
18 MARCH 1957
~~MIL-S-3562~~
~~9 AUGUST 1951~~

MILITARY SPECIFICATION

ADHESIVE, SEALING (FOR FILTERS)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers a liquid nitrile rubber base adhesive.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.** The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-282 — Filter Units, Protective Clothing, Gas-Mask Components and Related Products: Performance Test Methods.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS STANDARDS

D-553 — Viscosity and Total Solids Content of Rubber Cements.

D-816 — Testing Rubber Cements.

(Information as to the availability of American Society for Testing Materials Standards may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

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CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies of these freight classification rules should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 **Qualification.** The adhesive shall be a product which has been tested and passed the qualification tests specified in 4.5.1 (see 6.3).

3.2 **Materials.** The adhesive shall consist of nitrile rubber dissolved in acetone (see 4.2).

3.3 **Dioctylphthalate (DOP) penetration.** The adhesive shall not allow more than 0.05 percent DOP penetration when tested as specified in 4.5.1.1.

3.4 **Total solids content.** The total solids content shall be not less than 32 percent when tested as specified in 4.5.2.1.

3.5 **Viscosity.** The viscosity shall be not less than 1,500 nor more than 3,500 centipoises when tested as specified in 4.5.2.2.

3.6 **Softening point.** The softening point shall be not lower than 160°F. when tested as specified in 4.5.2.3.

3.7 **Cold flow.** The bond failure time shall be not less than 1 hour when tested as specified in 4.5.2.4.

3.8 **Cold brittleness.** The adhesive shall not crack or chip when tested as specified in 4.5.2.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 *Supplier's responsibility.* The supplier is responsible for the examination and testing as prescribed herein except for those examinations and tests reserved for performance by the Government.

4.1.2 *Government's responsibility.* The Government will accomplish the examinations and tests set forth in 4.5.1.

4.2 **Objective evidence.** The supplier shall present objective evidence as required by the Government representative that all materials are in accordance with 3.2 and section 5 (see 6.4).

4.3 **Alternate inspection (including testing) procedures.** The supplier may utilize any alternate inspection procedure which will assure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to its institution. In case of dispute, the procedures of the specification will govern.

4.4 **Inspection provisions.**

4.4.1 *Regular production lot.* A regular production lot shall consist of the adhesive produced by one manufacturer, in not more than 24 consecutive hours, under essentially the same manufacturing conditions and with no change in materials, provided the operation is continuous. In the event that the process is a batch operation, each batch shall constitute a lot (see 6.5).

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4.4.2 *Sampling.*

4.4.2.1 *Qualification test.* One gallon of adhesive, in four 1-quart containers, which is representative of the regular production of the supplier shall be submitted to a Government laboratory designated by the contracting officer for the qualification tests specified herein.

4.4.2.2 *Regular production.* A sample shall be taken from each lot, the size being calculated on the basis of one-tenth of the square root of the number of containers in the lot raised to the next higher whole number. A minimum of 1 quart shall be taken from each container unless the unit container is less than 1 quart in which case the contents of the unit shall be taken as a sample. In no case shall fewer than three containers be selected. If there are fewer than three containers in the lot each container shall be sampled. Equal portions of samples shall be composited. Tests shall be conducted on the composite sample in accordance with 4.5.2.

4.4.3 *Examination.* Each regular production lot shall be examined in accordance with the classification of defects and the AQL's as designated by this specification and with Standard MIL-STD-105.

4.4.4 *Classification of defects.*

4.4.4.1 *Preparation for delivery* (sec. 5).

Categories and defects

Major--AQL 4.0 percent defective.

1. Damaged containers.
2. Marking incorrect, missing or illegible.

4.5 **Tests.**

4.5.1 *Qualification.* The qualification tests shall consist of all the acceptance tests and the following additional tests. These tests are to be performed by the Government.

4.5.1.1 *DOP penetration.* The adhesive shall be used to fabricate twelve M13 particulate filters. The perforated sheets and gaskets shall be omitted. The filters shall be tested for DOP penetration in accordance with Method 102.1 of Standard MIL-STD-282. If three or more filters fail the DOP test, the adhesive shall be rejected. From the filters which passed the DOP penetration test, three filters shall be stored in a tropical chamber (113° ± 5°F. and 85 percent relative humidity); three in a desert chamber (160°F. min.); three in an arctic chamber (—65°F. min.) and the remainder shall be retained for control. Filters shall be stored for one week and immediately upon removal from the climatic chambers all filters shall be rough handled for 15 minutes in accordance with method 105.9 of Standard MIL-STD-282. After rough handling each filter shall be tested for DOP penetration in accordance with method 102.1 of Standard MIL-STD-282.

4.5.2 *Acceptance.* Tests for acceptance of reguar production lots shall be conducted as follows:

4.5.2.1 *Total solids.* Total solids shall be determined in accordance with ASTM method D553.

4.5.2.2 *Viscosity.* The viscosity shall be determined by using a model RVF Brookfield viscosimeter assembled with a number 3 spindle and operated at 20 revolutions per minute in accordance with the directions supplied by the manufacturer of the instrument. The adhesive shall be tested at a temperature of 70° ± 5°F. within 10 minutes after opening the sample container.

4.5.2.3 *Softening point.* Determine in accordance with ASTM method D816 that the softening point is not below 160°F. The sample drying time shall be from 5 to 20 minutes.

4.5.2.4 *Cold flow.* The bond failure time shall be determined in accordance with ASTM method D816 using an 8-pound load.

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The specimen drying time shall be from 5 to 20 minutes.

4.5.2.5 *Cold brittleness.* The cold brittleness shall be determined at —40°F. in accordance with ASTM method D816.

4.5.3 *Acceptance/rejection criteria.*

4.5.3.1 *Qualification.* If the adhesive fails any of the tests specified in 4.5.1, it shall be rejected.

4.5.3.2 *Rejection.* If the composite sample fails any of the tests specified in 4.5.2, the lot shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *Level A and B.* Not applicable.

5.1.2 *Level C.* The adhesive shall be packed in conformance with the manufacturer's commercial practice to provide protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers used shall comply with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract or order, all shipments shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. The material covered by this specification is intended for use as a sealing material in the manufacture of filters used in collective protectors.

6.2 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders, for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Commanding General, U. S. Army Chemical Center and Chemical Corps Materiel Command, Army Chemical Center, Md.

6.4 Objective evidence. Provisions for objective evidence and inspection records, and maintenance of inspection records will be specified by the procuring activity.

6.5 Batch. A batch is defined as the quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Chemical Corps
Navy—Bureau of Yards & Docks

Preparing activity:
Army—Chemical Corps

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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